Machine Learning Video Resources

The following StatQuest Machine Learning videos by Josh Starmer have been selected for their relevance to our course. You are expected to watch them to reinforce what you learn in class and as part of your homework. The videos are in English, with Turkish and English subtitles available.

Introduction to Machine Learning and Data Preparation

- A Gentle Introduction to Machine Learning
- One-Hot, Label, Target and K-Fold Target Encoding, Clearly Explained!!!

Regression and Optimization

- The Main Ideas of Fitting a Line to Data (The Main Ideas of Least Squares and Linear Regression.)
- Linear Regression, Clearly Explained!!!
- Multiple Regression, Clearly Explained!!!
- Gradient Descent, Step-by-Step
- Stochastic Gradient Descent, Clearly Explained!!!

Classification and Performance Evaluation

- StatQuest: Logistic Regression
- Logistic Regression Details Pt1: Coefficients
- Machine Learning Fundamentals: Confusion Matrix
- Machine Learning Fundamentals: Sensitivity and Specificity
- ROC and AUC, Clearly Explained!
- Neural Networks Part 6: Cross Entropy

Generalization and Regularization

- Machine Learning Fundamentals: Cross Validation
- Machine Learning Fundamentals: Bias and Variance
- Regularization Part 1: Ridge (L2) Regression
- Regularization Part 2: Lasso (L1) Regression
- Ridge vs Lasso Regression, Visualized!!!

Naive Bayes, kNN, and Decision Trees

- Naive Bayes, Clearly Explained!!!
- Gaussian Naive Bayes, Clearly Explained!!!
- StatQuest: K-nearest neighbors, Clearly Explained
- Entropy (for data science) Clearly Explained!!!
- Decision and Classification Trees, Clearly Explained!!!
- Regression Trees, Clearly Explained!!!

- How to Prune Regression Trees, Clearly Explained!!!
- Classification Trees in Python from Start to Finish

Ensemble Models

- StatQuest: Random Forests Part 1 - Building, Using and Evaluating
- AdaBoost, Clearly Explained
- Gradient Boost Part 1 (of 4): Regression Main Ideas
- Gradient Boost Part 2 (of 4): Regression Details
- Gradient Boost Part 3 (of 4): Classification
- Gradient Boost Part 4 (of 4): Classification Details
- XGBoost Part 1 (of 4): Regression
- XGBoost Part 2 (of 4): Classification
- XGBoost in Python from Start to Finish
- CatBoost Part 1: Ordered Target Encoding
- CatBoost Part 2: Building and Using Trees

Unsupervised Learning

- StatQuest: K-means clustering
- Clustering with DBSCAN, Clearly Explained!!!
- StatQuest: Principal Component Analysis (PCA), Step-by-Step
- StatQuest: The Main Ideas of PCA in Just 5 Minutes!!!
- StatQuest: PCA - Practical Tips
- StatQuest: PCA in Python

Deep Learning

- The Essential Main Ideas of Neural Networks
- Neural Networks Pt. 2: Backpropagation Main Ideas
- Neural Networks Pt. 3: ReLU In Action!!!
- Neural Networks Pt. 4: Multiple Inputs and Outputs
- Neural Networks Part 8: Image Classification with Convolutional Neural Networks (CNNs)
- Recurrent Neural Networks (RNNs), Clearly Explained!!!
- Long Short-Term Memory (LSTM), Clearly Explained
- Transformer Neural Networks, ChatGPT's foundation, Clearly Explained!!!
- The StatQuest Introduction to PyTorch